

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

Apr-May 2018

EC245/207.01/207 Control Systems

Date: 07.05.2018, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 The system with unity feedback having transfer function **[07]**

$$G(s) = \frac{128}{s(s^7 + 3s^6 + 10s^5 + 24s^4 + 48s^3 + 96s^2 + 128s + 192)}$$

Find total number of poles on left hand of s-plane also comment on stability.

Q - 2 Answer any two questions. **[14]**

I. A feedback Control System has an open loop transfer function

$$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Sketch the Root Locus as K is varied from 0 to ∞ .

II. Consider a unity feedback control system with an open loop transfer function

$$G(s)H(s) = \frac{K(s+1)(s+2)}{(s+0.1)(s-1)}$$

Draw the root locus of the system with the gain K as a variable.

III. A unity feedback control system has a n open –loop transfer function of

$$G(s) = \frac{K \left(s + \frac{4}{3} \right)}{s^2(s+12)}$$

Draw the Root Locus.

Q - 3 (a) Define the following terms. **[04]**

- 1) Transient Response
- 2) Damping Frequency
- 3) Steady State Error
- 4) Rise Time

Q - 3 (b) For a system $G(s)H(s) = \frac{K}{s^2(s+2)(s+3)}$, find the value of K to limit steady state error [05]

to 10 when input to system is $1 + 10t + \frac{40}{2}t^2$.

Q - 3 (c) A second order system has unity feedback and open loop transfer function [05]

$$G(s) = \frac{500}{s(s+15)}$$

- 1) What is characteristics equation?
- 2) What is Natural frequency of oscillations and damping ratio?
- 3) Calculate Peak time and Peak overshoot.
- 4) If input is ramp of 0.5 rad/sec, calculate steady state error.

OR

Q - 3 (a) Do as directed. [04]

- 1) Derivation of Peak Time
- 2) Derivation of Peak Overshoot

Q - 3 (b) A unity feedback system has [05]

$$G(s) = \frac{40(s+2)}{s(s+1)(s+4)}$$

Determine

- 1) Type of system
- 2) All error coefficients
- 3) Error for ramp input with magnitude 4.

Q - 3 (c) A unity feedback system has an open loop transfer function given below: [05]

$$G(s) = \frac{(1+0.4s)}{s(s+0.6)}$$

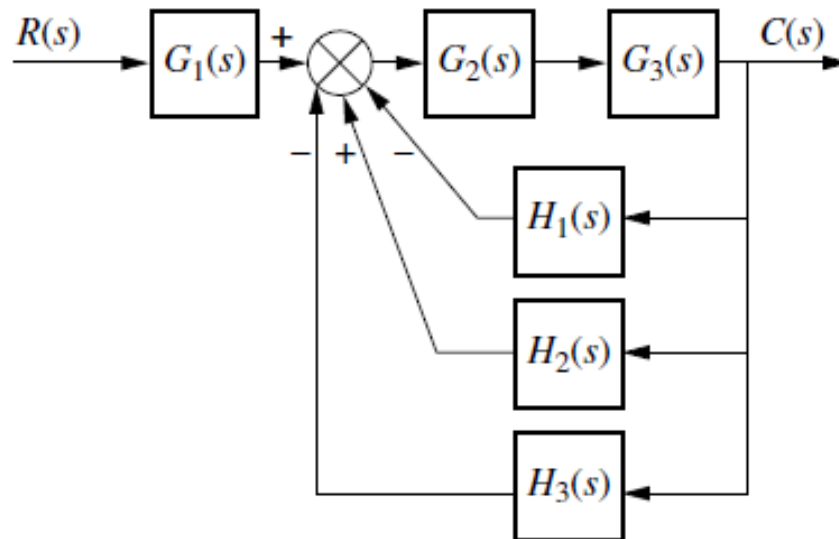
- 1) Obtain Unit step response.
- 2) What is Rise time?
- 3) What is peak overshoot?

SECTION – II

Q-4 (a) Give the differences between open loop control system and closed loop control systems in detail. [05]

Q-4 (b) Explain the importance of control system with any real world example. [02]

Q-5 (a) Reduce block diagram in canonical form of a system given below. [04]



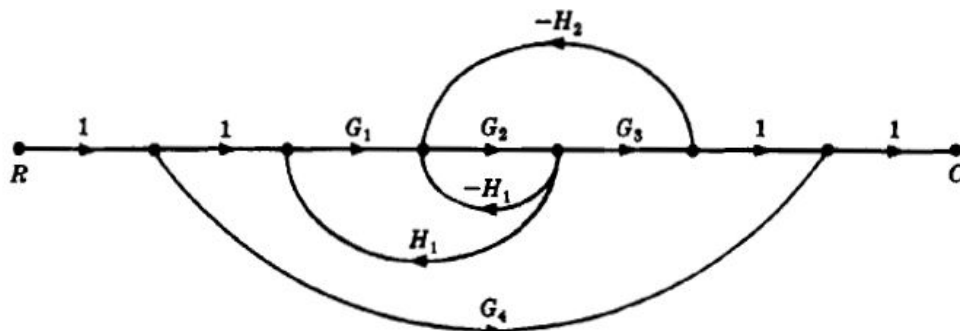
Q-5 (b) Answer any two questions.

[10]

- (i) I. Using Nyquist criterion, examine the closed loop stability of a system whose open loop transfer function is

$$G(s) = \frac{90}{s^2 + 9s + 18}$$

- II. Obtain transfer function from the signal flow graph shown in following figure (Use Mason's gain formula).



- III. Find the stability of the following system using Nyquist Criteria.

$$G(s) = \frac{90}{s^2 - 3s - 18}$$

Q - 6 (a) A unity feedback system has an open loop transfer function of

[07]

$$G(s) = \frac{100}{s(s+1)(s+2)}$$

Draw the bode plots and determine

- 1) Gain margin
- 2) Phase margin

- 3) Gain crossover frequency
- 4) Phase crossover frequency
- 5) State about the stability

OR

Q - 6 (a) Sketch the bode plot for the given transfer function,

[07]

$$G(s)H(s) = \frac{48(s+10)}{s(s+20)(s^2 + 2.4s + 16)}$$

Define the system stability.

Q - 6 (b) Discuss the effect on the system performance:

[07]

- 1) Proportional control
- 2) Integral control
- 3) Derivative control.

OR

Q - 6 (b) Draw the mechanical network for the mechanical system shown in figure below. [07]

Obtain an analogues electrical circuit based on Force- Current analogy.

